

L-O 1: Prepare Server Environment

Section 1:

Objective: Recall facts, concepts, and basic definitions.

1. Define the term **Windows Server**.
2. List any **three key components** of the Windows Server environment.
3. Identify **two types of operating systems** and give one example of each.
4. State any **two criteria** used for selecting an appropriate server.
5. Mention the **hardware requirements** necessary before installing Windows Server.

Section 2:

Objective: Explain ideas, interpret concepts, and summarize content.

1. Explain the difference between **client-side** and **server-side** operating systems.
2. Describe the role of **Hyper-V** in Windows Server.
3. Differentiate between **standalone OS** and **network OS**.
4. Summarize the importance of **virtualization** in modern server environments.
5. Explain why **UPS** and **rack enclosures** are important in a server setup.

Section 3:

Objective: Use knowledge in practical contexts or real-world scenarios.

1. Demonstrate how you would **configure a network adapter** for a Windows Server.
2. Apply the correct procedure to **create installation media** for a Windows Server setup.
3. Show how to **assign a static IP address** on a Windows Server.
4. Install the **necessary drivers and services** after a fresh Windows Server installation.
5. Identify and configure **firewall settings** to allow secure network connectivity.

Section 4:

Objective: Break down information, distinguish components, and identify relationships.

1. Compare **physical and virtual installation** of Windows Server, outlining the pros and cons of each.
2. Analyze how **hardware components (CPU, RAM, storage)** influence server performance.
3. Differentiate between **Server Core** and **Server Manager** installations.
4. Examine how **network sharing and discovery settings** affect server accessibility.
5. Analyze the relationship between **server selection criteria** and **expected workload**.

Section 5:

Objective: Make judgments, justify decisions, and critique alternatives.

1. Evaluate the importance of **virtualization software** when deploying multiple servers.
2. Justify your choice of **Windows Server edition** for a small vs. large organization.
3. Assess the impact of **improper IP configuration** on server performance.
4. Judge the effectiveness of **server management tools** in maintaining uptime.
5. Critically evaluate how **firewall and security settings** balance protection and accessibility.

Section 6:

Objective: Design, plan, or construct a new solution or configuration.

1. Design a **server environment setup plan** for a medium-sized enterprise, considering hardware, OS type, and virtualization.
2. Create a **step-by-step installation and configuration guide** for Windows Server on a virtual machine.
3. Develop a **checklist** for preparing and verifying all **server tools and equipment** before installation.

4. Propose a **network configuration model** that ensures optimal connectivity and security for Windows Server clients.
5. Construct a **maintenance strategy** for keeping Windows Server drivers and services up to date.

Section 7: Scenario-Based Task

Scenario:

Padri Vjeko Centre TSS, a technical and vocational school with over **500 students and 30 trainers**, offers training across four departments — **NIT (Networking and Information Technology)**, **FAD (Fashion and Design)**, **WOT (Wood Technology)**, and **BDC (Building and Construction)**.

To enhance performance, centralize resource management, and improve data security and user control, the school plans to **migrate its academic and administrative systems to Windows Server**. The new server environment will host essential services such as the **Student Management System, Attendance Tracking, Centralized File Sharing, Departmental Web Projects, and Local Intranet Access** for trainers and students.

Task:

As the **System Administrator** of Padri Vjeko Centre TSS, outline the **steps and considerations** you would take to **install and configure a Windows Server environment** to effectively support the school's operations.